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R Notebook: EEG Frontal Asymmetry (FA)

This Help Center article describes the EEG Frontal Asymmetry and EEG Frontal Asymmetry Aggregation R Notebooks for EEG.

The Frontal Asymmetry (FA) R Notebook runs a complete signal processing protocol for obtaining FA scores across different power bands. The FA scores can be used as a EEG-based metrics for further statistical analysis or feature extraction.

Note that FA is an example of an EEG-based metric that is derived from a Power Spectral Decomposition of the EEG signal (see also: R Notebook (EEG): Power Spectral Density).

This Help Center article will guide you through:

- What is Frontal Asymmetry and Frontal Alpha Asymmetry (FAA)?
- How to add the FA R Notebook to your analysis
- Details about the algorithm
- Parameters
- Is it possible to use the ABM 'decontaminated' data?
- Is it possible to use the power band data from Emotiv?
- Is it possible to run EEG R Notebooks on data collected from a device connected via LSL?
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What is Frontal Asymmetry

Frontal Asymmetry is a measure of the difference in brain activity over the left and right frontal regions of the brain (Briesemeister et al., 2013). The difference is usually calculated for a specific frequency band and in iMotions computations are available for all commonly used frequency bands: delta, theta, alpha, beta and gamma frequency bands. A commonly used metric is that of frontal alpha (8-12Hz) asymmetry described in the next section.

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What is Frontal Alpha Asymmetry (FAA)

Frontal Alpha Asymmetry (FAA) is an EEG-based metric that is related to ‘approach and withdrawal related motivation’: two fundamental behavioral tendencies that govern how individuals respond to internal or external stimuli in their environment. Frontal asymmetry is actively used in a great variety of disciplines within applied psychology. This includes, among others, mental health, product design, communications and marketing research. While approach motivation is commonly associated with positive emotions such as joy or enthusiasm, it can also be triggered by negative emotions like anger, which can propel action toward overcoming obstacles or confronting perceived threats. In contrast, withdrawal motivation is characterized by the urge to retract or avoid. Importantly, these motivational systems operate independently of valence, meaning that approach behaviors are not exclusively linked to positive emotions, nor are withdrawal behaviors solely tied to negative emotions; rather, they reflect distinct motivational directions driven by both emotional and situational factors.

> Note:
>
> Alpha power is inversely related to the level of brain activity in the alpha range; a decrease in alpha power reflects an increase in brain activity. This is due to the synchronization of alpha waves during states of rest, and desynchronization of alpha waves during states of activity.

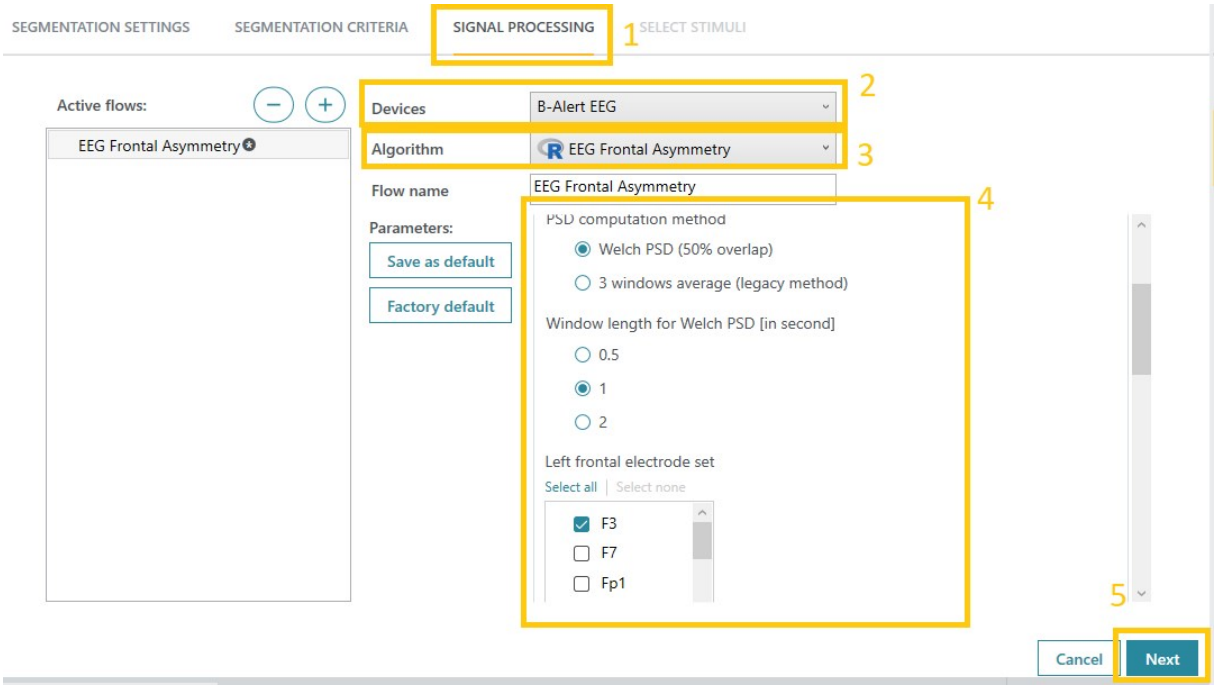
How to add the FA R Notebook to your analysis

The Frontal Asymmetry (FA) R Notebook can be added to an analysis during the creation process of a new analysis, or later, by returning to the signal processing page within an existing analysis.

To add the R Notebook to the analysis:

1. Navigate to the Signal Processing page, and click the '+' button to add an active flow
2. In the 'Devices' drop-down menu, select the correct EEG device
3. In the 'Algorithms' drop-down menu, select the EEG Frontal Asymmetry R Notebook
4. Set the parameters (see also: Parameters)
5. Click 'Next'

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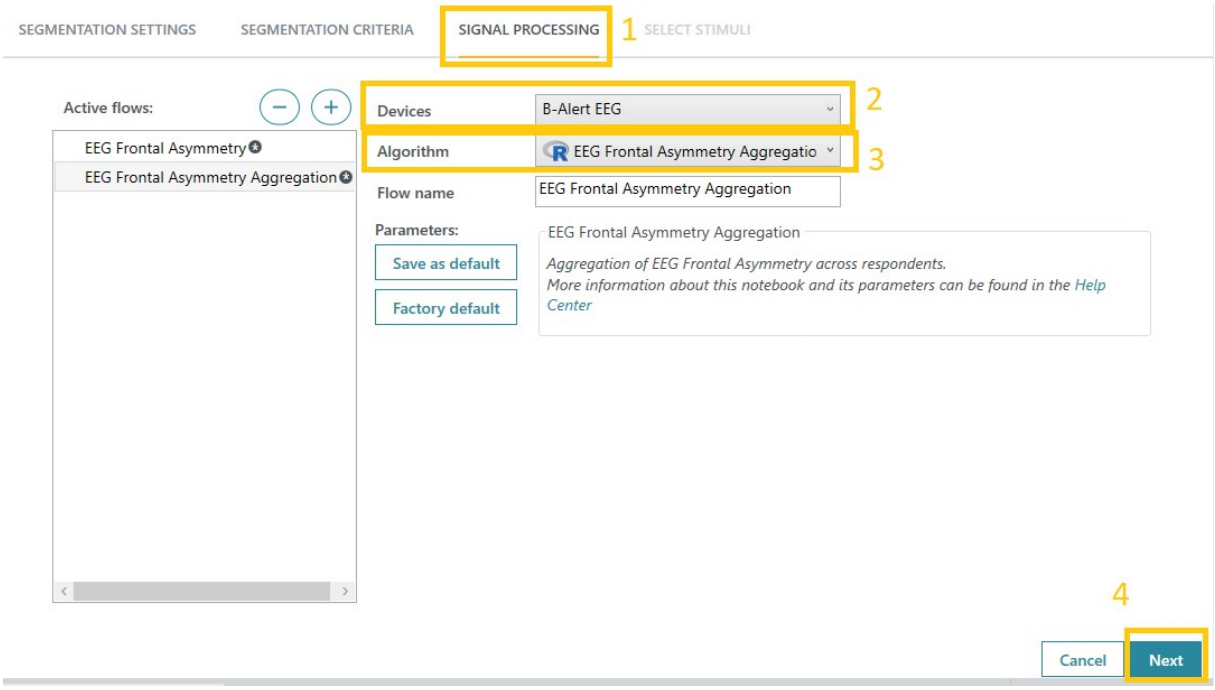
If you also wish to compute an 'aggregation' of the FA scores (i.e., the average across all respondents in your analysis), you need to add the FA aggregation R Notebook.

To add the FA aggregation R Notebook to the analysis:

1. At the Signal Processing page, click the '+' button to add another active flow
2. In the 'Devices' drop-down menu, select the EEG device
3. In the 'Algorithms' drop-down menu, select the EEG Frontal Asymmetry Aggregation R Notebook
4. Click 'Next'

> **Note:**
>
> In order to run EEG Frontal Asymmetry Aggregation R Notebook, you have to run EEG Frontal Asymmetry R Notebook on an individual level first.

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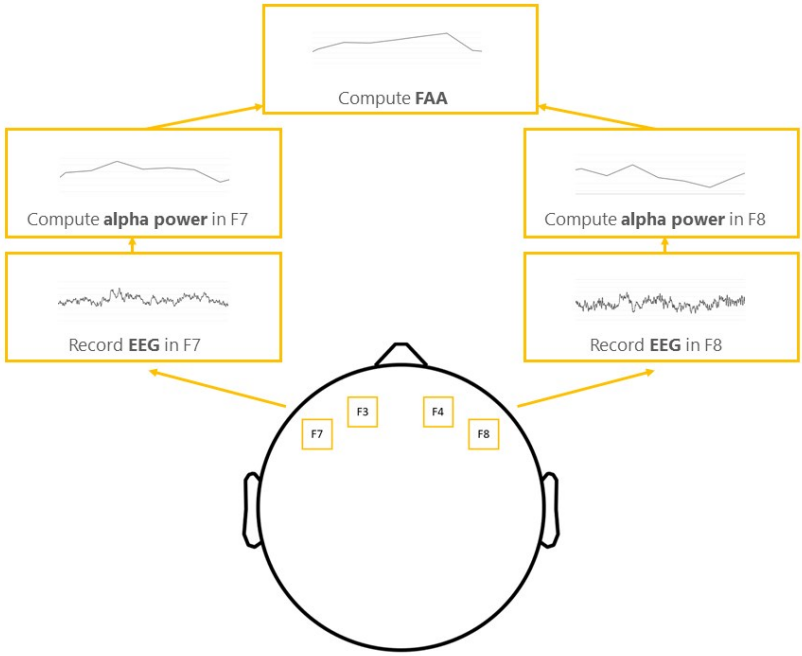
Details about the algorithm

The Frontal Asymmetry (FA) index reflects difference in power (or 'asymmetry') between two sets of electrodes placed over the left and right frontal hemispheres, respectively. Often electrode locations F7 and F8 are used – alternatively F3 and F4 or a average of F3/F7 and F4/F8 (depending on the available electrode locations in the EEG system used and your research method).

The available power bands in iMotions for computation are: Delta (default range: 1 - 3 Hz), Theta (4 - 7 Hz), Alpha (8 - 12 Hz), Beta (13 - 25 Hz), Gamma (26 - 40 Hz).

Read more: How to place the EEG-system correctly on the head

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The R Notebook performs a Power Spectral Decomposition of the EEG signals in the following steps:

1. Pre-processing of the signal: bandpass filtering (a zero phase-lag band-pass butterworth filter); notch filtering (a zero phase-lag butterworth notch filter); re-referencing (for BrainProducts only); artifact rejection: rejecting data-points whose absolute values exceed a set voltage threshold
2. Fast Fourier Transform (FFT) of the EEG signal with window sizes of 1-second (default) or 0.5-second or 2-second time windows as optional settings.
3. Average PSD within a certain frequency range.

> **More Information:**
>
> Please refer to the PSD R Notebook article for more descriptions on topics: "What is Power Spectral Density" and " What is the difference between our legacy method and the Welch method"

4. Calculation of asymmetry scores.

Using Frontal Asymmetry Alpha as an example, the index is computed by the following formula:

$$\mathbf{FAA} = \ln\left(\frac{Alpha_{F8}}{Alpha_{F7}}\right)$$

Or equivalently:

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$$\mathbf{FAA} = \ln\left(Alpha_{F8}\right) - \ln\left(Alpha_{F7}\right)$$

The F8 and F7 electrodes are just used as examples here and could be replaced with F4 and F3 or an average across several electrodes places on either hemisphere.

> Note:
>
> Any averaging across electrodes, time and participants is performed before applying any logarithmic transform to the data.

In the above example the motivation index is computed by subtracting the natural logarithm transformed alpha power in the left hemisphere from the natural logarithm transformed alpha power in the right hemisphere. If the alpha power is equal in the left and right hemisphere, the index is 0. But if the alpha power is relatively stronger in the right hemisphere, the index will become positive (i.e., the values are above 0), and vice versa. Positive scores signify 'approach motivation' (liking) and negative scores signify 'withdrawal motivation' (disliking).

Note that the frontal asymmetry scores should be interpreted relative to each other. For example, by comparing the index for two different stimuli: A and B. If the index is more positive for A than for B the conclusion would be that A evoked a stronger approach motivation than B. Or equivalently, A evoked less avoidance motivation than B.

Parameters

The R Notebook can be altered by changing the following parameters. Most parameters concern the pre-processing of the EEG data (e.g., filtering, artifact rejection) and the method used for computing power (e.g., Welch method, Powerbands options); therefore, most of these parameters are the same as for the Power Spectral Density (PSD) R Notebook.

Power calculation method:

- PSD computation method: Choose to run the ‘legacy method’ or ‘Welch method’ (standard method in the scientific literature).
- Window length for Welch PSD [in second]: Window length used for applying Fast Fourier Transform.

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Notes on Window length parameter:

1. Welch PSD method only. For legacy method a window size of 1 second will always be used.
2. The chosen window size affects the frequency resolution i.e. the number of frequency bins reported. Only change this parameter if you know what you are doing.

Available options and defaults of each parameter:

PSD computation method	"3 windows average (legacy method)" and "Welch PSD (50% overlap)". Default is "Welch PSD (50% overlap)"
Window length for Welch PSD [in second]	0.5, 1, 2. Default is 1 second

Electrode locations:

- Left frontal electrode set: Set one or more electrodes located on the left hemisphere for the calculation of left side powerbands activity
- Right frontal electrode set: Set one or more electrodes located on the right hemisphere for the calculation of right side powerbands activity
- LSL Left frontal electrode set (comma-separated): Only applicable to LSL devices. Input one or more electrodes located on the left hemisphere for the calculation of left side powerbands activity. Use "," to separate channels. Case-insensitive. Please note that manually providing the LSL channel names will now work even if the user forgot to create a profile / created the wrong profile for their device.
- LSL Right frontal electrode set (comma-separated): Only applicable to LSL devices. Input one or more electrodes located on the left hemisphere for the calculation of left side powerbands activity. Use "," to separate channels. Case-insensitive. Please note that manually providing the LSL channel names will now work even if the user forgot to create a profile / created the wrong profile for their device.

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Notes on Electrode locations:

Please note that the naming of electrode locations is different for Enobio than other EEG headsets. For Enobio, channel names are reported as Ch1, Ch2, Ch3... You will have to find the original channel names in the Enobio software.

Available options and defaults of each parameter:

Left frontal electrode set	Default is Ch1 for Enobio, F3 for other EEG headset
Right frontal electrode set	Default is Ch7 for Enobio, F4 for other EEG headset
LSL Left frontal electrode set (comma-separated)	Default is blank
LSL Right frontal electrode set (comma-separated)	Default is blank

Pre-processing:

Band-pass filter frequencies [in Hz]: The band-pass filter is used to preserve power within a specified frequency range.

Notch filter frequency: The Notch filter is used to remove line noise introduced in the signal by electrical equipment in the recording environment. In the US: 60 Hz in Europe: 50 Hz

Notes on Band-pass filter frequencies:

The lowpass cutoff frequency (by default 100 Hz) should be set lower than Nyquist frequency (i.e., 0.5 * sampling rate). If it exceeds the Nyquist frequency, the algorithm will use Nyquist frequency as the actual lowpass cutoff frequency.

Notes on Notch filter frequency:

Make sure you choose the right Notch filter frequency. If you choose the incorrect Notch filter frequency and set your bandpass frequency range big, it could result in line noise being kept in EEG signal. This may give you a lot of NA values later because of the subsequent artifact rejection.

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Reference channel (BrainProducts only): The re-referencing option allows for re-referencing the EEG signal to a custom channel (e.g., Cz, Mastoid(s), tip of the nose).

Artefact threshold [in µV]: The artifact threshold is a simple voltage threshold that marks the parts of the EEG signal where the absolute value of the data exceeds the threshold. These parts of the EEG signal will be excluded according to the method specified below. Note that EEG signals would typically fluctuate within a range of -100 to 100 microvolts around the mean (larger fluctuations may reflect artifacts).

Available options and defaults of each parameter:

Band-pass filter frequencies [in Hz]	Default range: 0.5 - 100 Hz Note: Please always use "-" as a separator
Notch filter frequency	50 Hz, 60 Hz. Default is 60 Hz
Artefact threshold [in µV]	Default is 120 µV
*Reference channel 1	Default is: 'Ref'
*Reference channel 2, leave empty for re-referencing based on only one electrode	Default is empty

* only available for BrainProducts (ActiChamp)

Powerbands:

- Powerbands options: The available powerbands for computation. Please note that you can choose one or multiple powerbands in order to get FA results from single or multiple powerbands.
- Delta band frequencies [in Hz]: Frequency range for delta band
- Theta band frequencies [in Hz]: Frequency range for theta band
- Alpha band frequencies [in Hz]: Frequency range for alpha band
- Beta band frequencies [in Hz]: Frequency range for beta band
- Gamma band frequencies [in Hz]: Frequency range for gamma band

Available options and defaults of each parameter:

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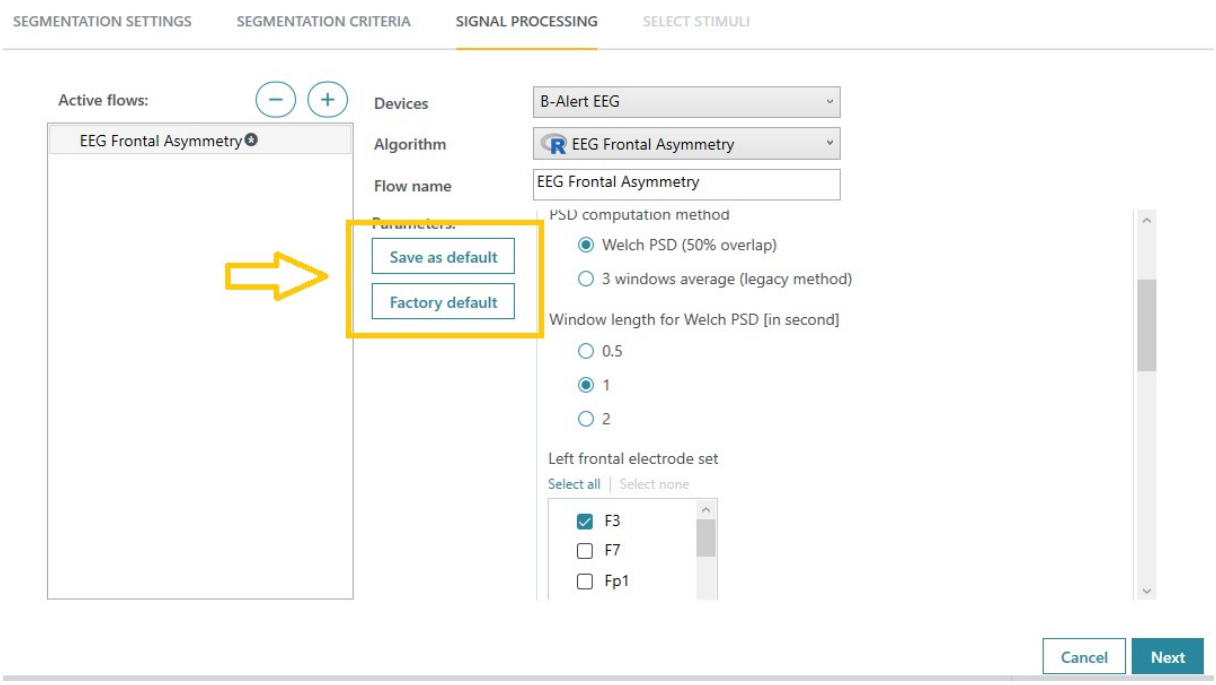
Powerbands options	Delta, Theta, Alpha, Beta, Gamma. Default is Alpha
Delta band frequencies [in Hz]	Default range: 1 - 3 Hz
Theta band frequencies [in Hz]	Default range: 4 - 7 Hz
Alpha band frequencies [in Hz]	Default range: 8 - 12 Hz
Beta band frequencies [in Hz]	Default range: 13 - 25 Hz
Gamma band frequencies [in Hz]	Default range: 26 - 40 Hz

Note:

Please always use "-" as a separator

Tip:

It is possible to change these default values and save them for later, by clicking ‘Save as default’ on the Signal processing page; at any time, you can restore the default values by clicking ‘Factory defaults’.



Is it possible to use the ABM ‘decontaminated’ data?

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Users of the ABM system may wonder whether it is possible to use the ‘decontaminated’ data provided by ABM, rather than the ‘raw’ data. The answer is No. You can only apply R Notebook on ABM raw data. Please make sure that you select the ABM 'raw' data (B-Alert EEG) instead of ABM 'decontaminated' data (B-Alert Decontaminated EEG) in the drop-down menu under 'Devices' (See also: How to add the FA R Notebook to your analysis). If you choose the ABM 'decontaminated' data (B-Alert Decontaminated EEG), there will be no available EEG R Notebooks shown.

Is it possible to use the ‘power bands‘ data from Emotiv?

Users of the Emotiv system may note that Emotiv already provides power band information directly. The most salient difference between the power band information computed by Emotiv and by iMotions, is that slightly different frequency bands are used: theta (4-8 Hz), alpha (8-12 Hz), low beta (12-16 Hz), high beta (16-25 Hz) and gamma (25-45 Hz). However, the algorithms developed by Emotiv are proprietary, and therefore, it can be assumed that the methods deviate from the methods described in this Help Center article. Users of the Emotiv system are able to choose to run the EEG Power Spectral Density R Notebook from the ‘raw’ EEG data, or from the 'power band' EEG data provided by Emotiv directly. This is done by either selecting Emotiv 'raw' data or Emotiv 'powerbands' data in the drop-down menu Devices (See also: How to add the FA R Notebook to your analysis).

If you choose to run the Frontal Asymmetry (FA) R Notebook from the raw EEG data, the entire computation happens exactly as described in this Help Center article.

If you choose to run the Frontal Asymmetry (FA) R Notebook from the power bands information provided by Emotiv, the R Notebook merely computes the natural logarithm of the power delivered by Emotiv. Hereafter, the R Notebook computes the FA index using the formula provided in this Help Center article.

Is it possible to run EEG R Notebooks on data collected from a device connected via LSL?

It is possible now since iMotions version 9.1.6. With the help of "Adding a Device profile" before your data collection, it is possible to run EEG R Notebooks on your collected data. Please refer to this article to learn more about LSL Device profiles and how to choose one before data collection.

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What the R Notebook delivers: replay, export files, report

The Frontal Asymmetry (FA) R Notebook computes:

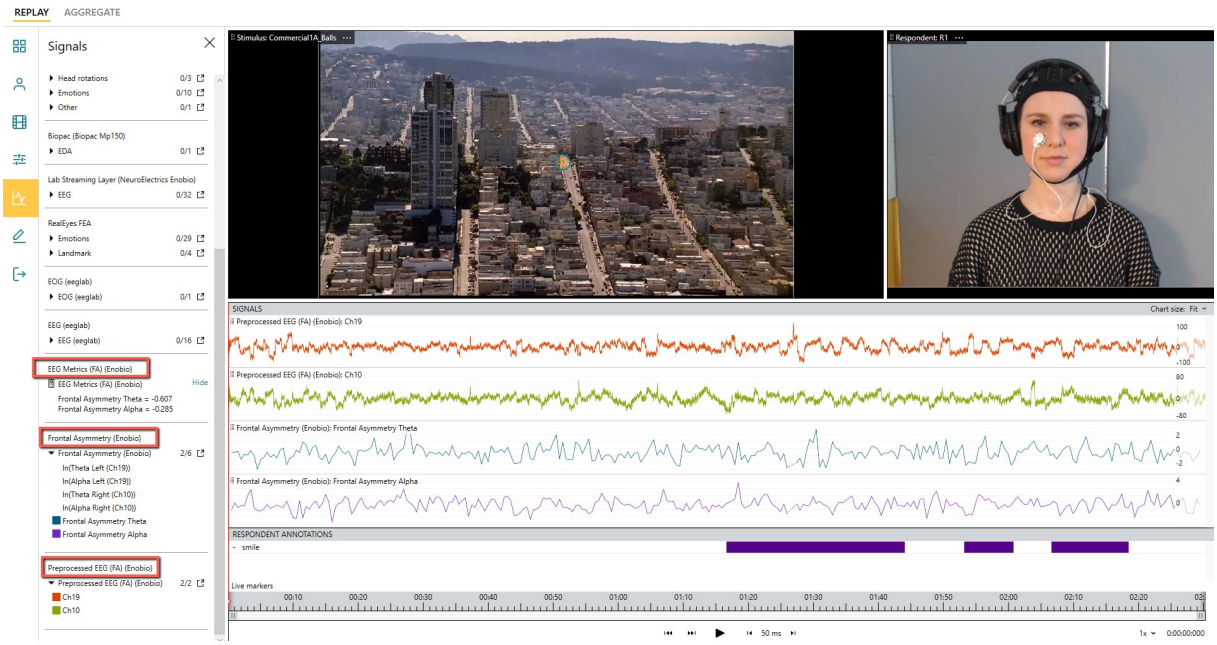
1. The pre-processed (band-pass filtered, notch filtered, re-referenced (Actichamp only), artifact rejected) EEG signal
2. The natural log transformed powerband power for left frontal electrode set, the natural log transformed powerband power for right frontal electrode set, and the FA score as timeline data
3. The mean FA for per stimulus, scene and annotation placed in the analysis
   These data can be visualized in the Individual Replay, Aggregate Replay and exported as part of the sensor data export file or as a separate summary metrics file.
4. A one-page report with graphs, algorithm descriptions and embedded R-code.

Individual Replay

The R Notebook adds three sections to the signals page in the Replay: 'Preprocessed EEG (FA)', ‘Frontal Asymmetry’ and ‘EEG Metrics (FA)’:

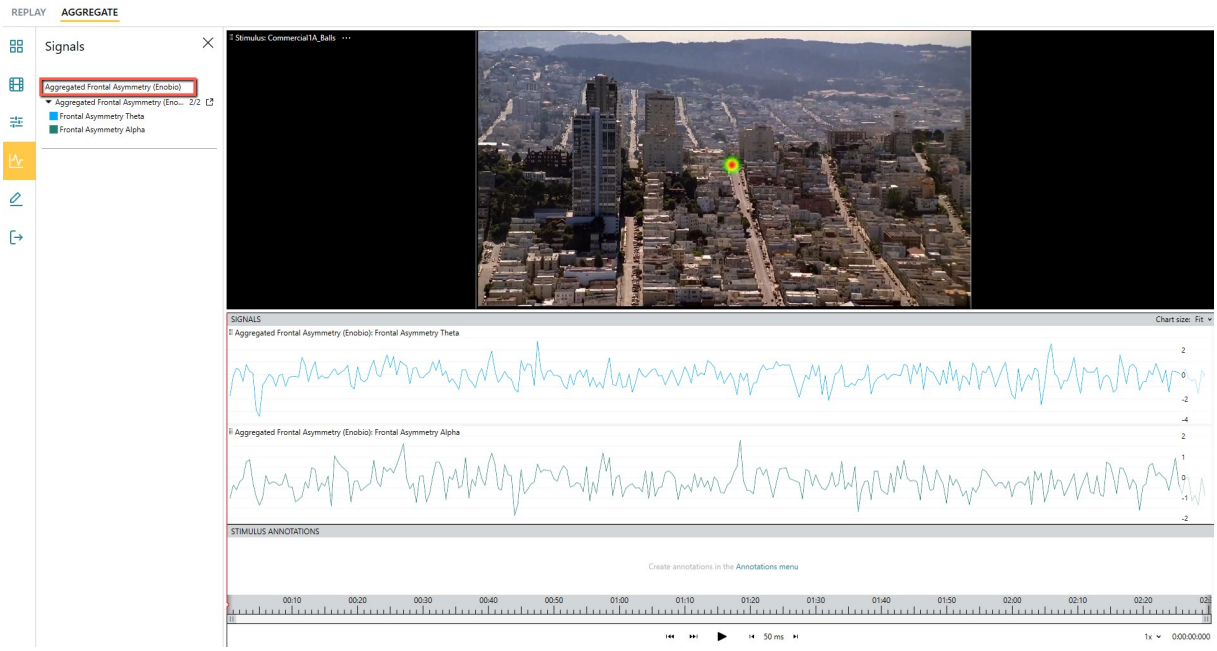
1. The section 'Preprocessed EEG (FA)' returns the pre-processed EEG signal of each selected channel.
2. The section ‘Frontal Asymmetry’ returns multiple data channels: the natural log transformed power of selected powerband(s) for left frontal electrode set (E.g., ln (F7) alpha), the natural log transformed power of selected powerband(s) for right frontal electrode set (E.g., ln (F8) alpha), and the FA score (E.g., Frontal Asymmetry Alpha, Frontal Asymmetry Theta). Note that results from multiple powerbands selected in one analysis will be displayed at the same time.
3. The section ‘EEG Metrics (FA)’ further provides a quick summary statistics for each selected powerband(s), namely, the mean FA for the respondent and stimulus currently displayed (Frontal Asymmetry Alpha, Frontal Asymmetry Theta). Note that, here, only the FA for the entire stimulus duration is calculated and displayed.

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Aggregate replay

If you choose to also add the FA aggregation R Notebook, in the Aggregate Replay you may find the section 'Aggregated Frontal Asymmetry'. This section contains multiple new data-channel, representing the aggregated FA scores per time unit (across all respondents included in the analysis) of each selected powerbands (E.g., Frontal Asymmetry Theta, Frontal Asymmetry Alpha).

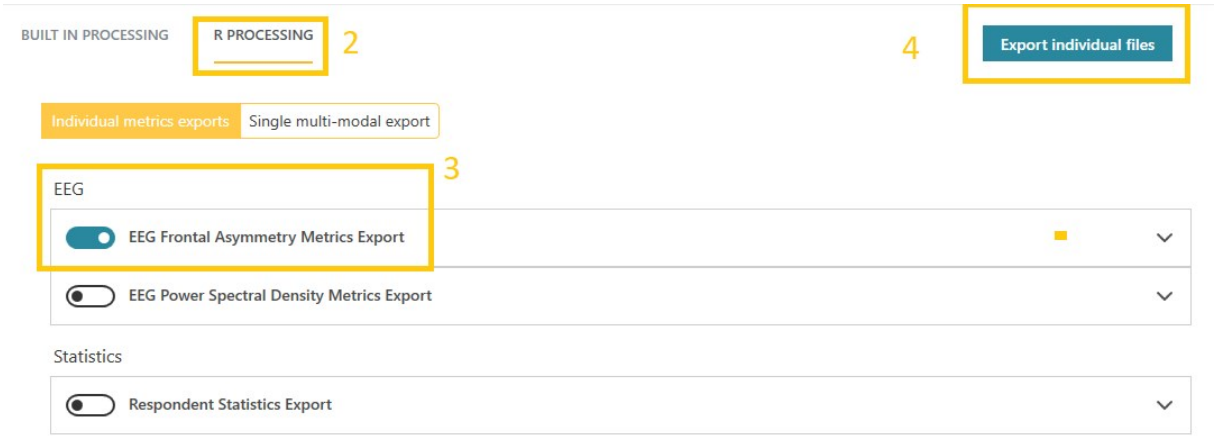


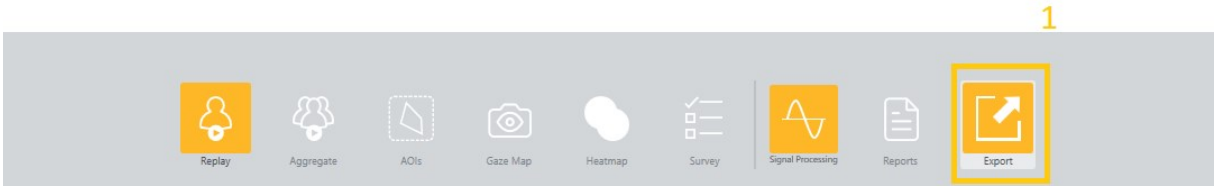
Metrics export

The R Notebook further produces an export file, which, for every respondent included in the analysis and for every stimulus, scene, or annotation presented in the study, computes the mean FA (E.g., Mean Frontal Asymmetry Theta, Mean Frontal Asymmetry Alpha).

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Please see here for a detailed description on the exported metrics summary file.



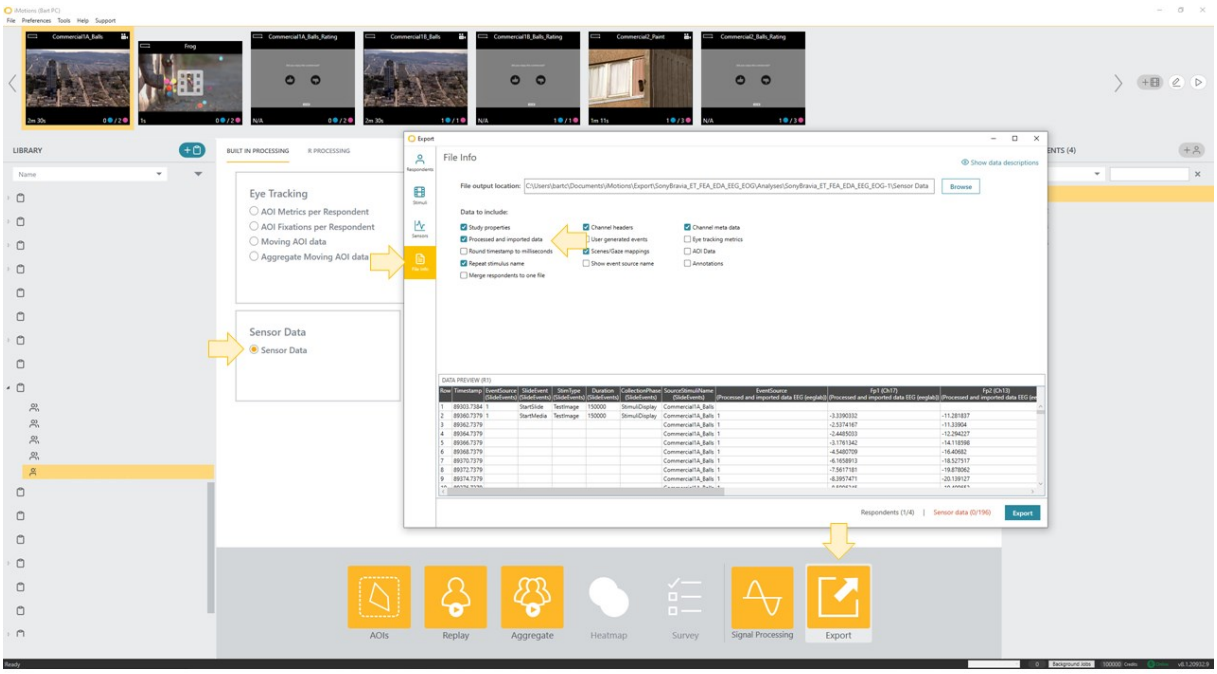


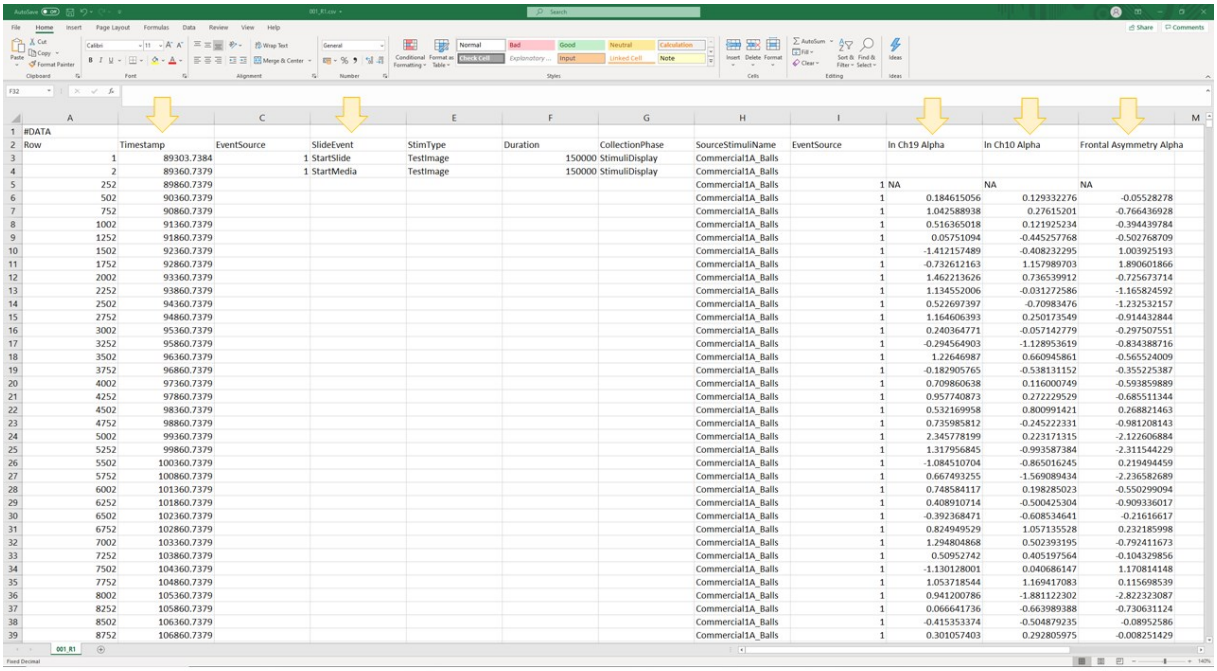
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Study Nan	Responde	Gender	Age	Group	Type	Label	Start (ms)	Duration (	Parent Sti	Mean Frontal Asymmetry Theta	Mean Frontal Asymmetry Alpha	Algorithm Device Name			
2	SonyBravi	R1	Female	27	Default	Stimulus	Commerci	89360	150341		-0.607	-0.285	Frontal Asymmetry (Enobio)			
3	SonyBravi	R1	Female	27	Default	Annotatio	Bin	89361	150000		-0.607	-0.285	Frontal Asymmetry (Enobio)			
4	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	89361	5000	Commerci	-0.843	-0.45	Frontal Asymmetry (Enobio)			
5	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	94361	5000	Commerci	-0.789	-0.887	Frontal Asymmetry (Enobio)			
6	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	99361	5000	Commerci	-0.333	-0.565	Frontal Asymmetry (Enobio)			
7	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	104361	5000	Commerci	-0.839	-0.407	Frontal Asymmetry (Enobio)			
8	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	109361	5000	Commerci	-0.131	-0.077	Frontal Asymmetry (Enobio)			
9	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	114361	5000	Commerci	0.304	0.422	Frontal Asymmetry (Enobio)			
10	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	119361	5000	Commerci	-0.67	0.051	Frontal Asymmetry (Enobio)			
11	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	124361	5000	Commerci	-0.751	-0.289	Frontal Asymmetry (Enobio)			
12	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	129361	5000	Commerci	-1.019	-0.394	Frontal Asymmetry (Enobio)			
13	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	134361	5000	Commerci	-0.078	-0.428	Frontal Asymmetry (Enobio)			
14	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	139361	5000	Commerci	-0.063	-0.571	Frontal Asymmetry (Enobio)			
15	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	144361	5000	Commerci	-0.993	0.192	Frontal Asymmetry (Enobio)			
16	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	149361	5000	Commerci	-1.033	-0.561	Frontal Asymmetry (Enobio)			
17	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	154361	5000	Commerci	-0.819	-0.403	Frontal Asymmetry (Enobio)			
18	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	159361	5000	Commerci	-0.02	-0.589	Frontal Asymmetry (Enobio)			
19	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	164361	5000	Commerci	-1.114	0.22	Frontal Asymmetry (Enobio)			
20	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	169361	5000	Commerci	-0.509	0.147	Frontal Asymmetry (Enobio)			
21	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	174361	5000	Commerci	-0.79	-0.274	Frontal Asymmetry (Enobio)			
22	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	179361	5000	Commerci	-0.78	-0.209	Frontal Asymmetry (Enobio)			
23	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	184361	5000	Commerci	-0.401	-0.114	Frontal Asymmetry (Enobio)			
24	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	189361	5000	Commerci	-0.47	-0.449	Frontal Asymmetry (Enobio)			
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26	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	199361	5000	Commerci	-1.373	-0.751	Frontal Asymmetry (Enobio)			
27	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	204361	5000	Commerci	-0.26	0.062	Frontal Asymmetry (Enobio)			
28	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	209361	5000	Commerci	0.223	0.084	Frontal Asymmetry (Enobio)			
29	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	214361	5000	Commerci	-0.442	-0.771	Frontal Asymmetry (Enobio)			
30	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	219361	5000	Commerci	-0.619	-0.042	Frontal Asymmetry (Enobio)			
31	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	224361	5000	Commerci	-0.339	-0.444	Frontal Asymmetry (Enobio)			
32	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	229361	5000	Commerci	-0.934	0.206	Frontal Asymmetry (Enobio)			
33	SonyBravi	R1	Female	27	Default	Annotatio	Bin (instar	234361	5000	Commerci	-0.785	-0.563	Frontal Asymmetry (Enobio)			
34	SonyBravi	R1	Female	27	Default	Annotatio	Scenes	89361	150083		-0.607	-0.285	Frontal Asymmetry (Enobio)			
35	SonyBravi	R1	Female	27	Default	Annotatio	Scenes (in	89361	2333	Commerci	-0.658	-0.656	Frontal Asymmetry (Enobio)			
36	SonyBravi	R1	Female	27	Default	Annotatio	Scenes (in	91694	1458	Commerci	-0.539	0.923	Frontal Asymmetry (Enobio)			
37	SonyBravi	R1	Female	27	Default	Annotatio	Scenes (in	93152	1375	Commerci	-1.39	-0.951	Frontal Asymmetry (Enobio)			

Sensor data export

It is also possible to export the FA data as part of the sensor data export file. This is done by ticking the option: ‘Processed and imported data’ in the sensor data export editor.

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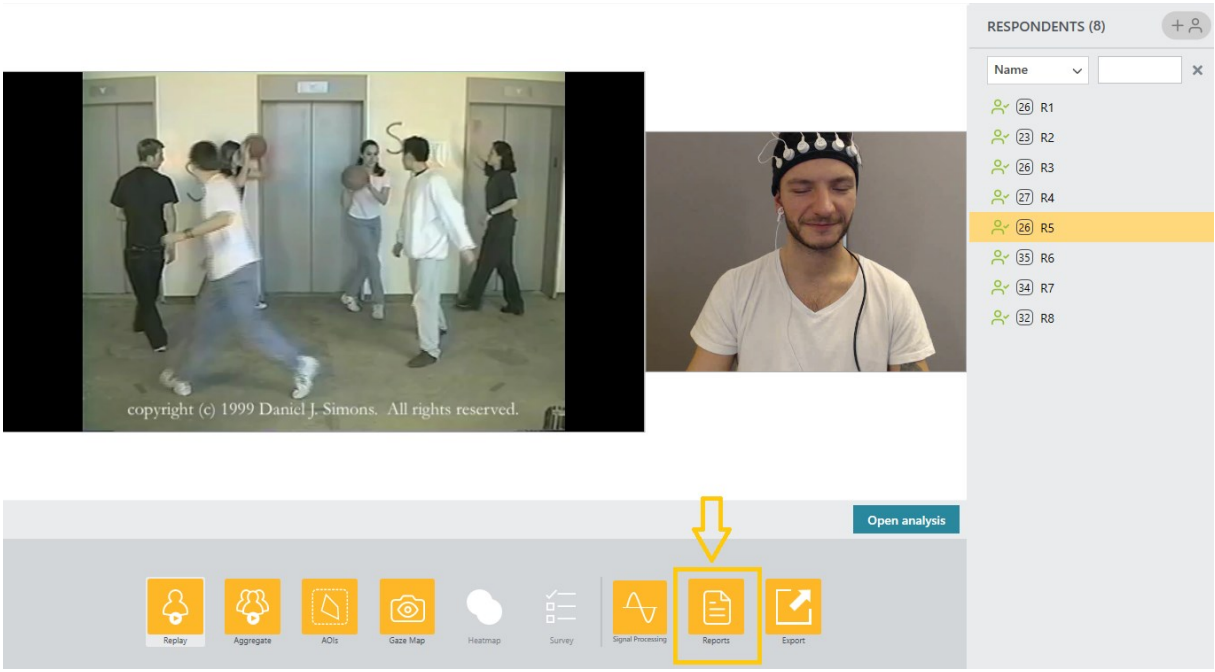
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	#DATA												
2	Row	Timestamp	EventSource	SlideEvent	StimType	Duration	CollectionPhase	SourceStimuliName	EventSource	ln Ch19 Alpha	ln Ch10 Alpha	Frontal Asymmetry Alpha	
3	1	89303.7384		1 StartSlide	TestImage	150000	StimuliDisplay	Commercial1A_Balls					
4	2	89360.7379		1 StartMedia	TestImage	150000	StimuliDisplay	Commercial1A_Balls					
5	252	89860.7379						Commercial1A_Balls	1	NA	NA	NA	
6	502	90360.7379						Commercial1A_Balls	1	0.184615056	0.129332276	-0.05528278	
7	752	90860.7379						Commercial1A_Balls	1	1.042588938	0.27615201	-0.766436928	
8	1002	91360.7379						Commercial1A_Balls	1	0.516365018	0.121925234	-0.394439784	
9	1252	91860.7379						Commercial1A_Balls	1	0.05751094	-0.445257768	-0.502768709	
10	1502	92360.7379						Commercial1A_Balls	1	-1.412157489	-0.408232295	1.003925193	
11	1752	92860.7379						Commercial1A_Balls	1	-0.732612163	1.157989703	1.890601866	
12	2002	93360.7379						Commercial1A_Balls	1	1.462213626	0.736539912	-0.725637314	
13	2252	93860.7379						Commercial1A_Balls	1	1.134552006	-0.031272586	-1.165824592	
14	2502	94360.7379						Commercial1A_Balls	1	0.522697397	-0.70983476	-1.232532157	
15	2752	94860.7379						Commercial1A_Balls	1	1.164606393	0.250173549	-0.914432844	
16	3002	95360.7379						Commercial1A_Balls	1	0.240364771	-0.057142779	-0.297507551	
17	3252	95860.7379						Commercial1A_Balls	1	-0.294564903	-1.128953619	-0.834388716	
18	3502	96360.7379						Commercial1A_Balls	1	1.22646987	0.660945861	-0.565524009	
19	3752	96860.7379						Commercial1A_Balls	1	-0.182905765	-0.538131152	-0.355225387	
20	4002	97360.7379						Commercial1A_Balls	1	0.709860638	0.116000749	-0.593859889	
21	4252	97860.7379						Commercial1A_Balls	1	0.957740873	0.272229529	-0.685511344	
22	4502	98360.7379						Commercial1A_Balls	1	0.532169958	0.800991421	0.268821463	
23	4752	98860.7379						Commercial1A_Balls	1	0.735985812	-0.245222331	-0.981208143	
24	5002	99360.7379						Commercial1A_Balls	1	2.345778199	0.223171315	-2.122606884	
25	5252	99860.7379						Commercial1A_Balls	1	1.317956845	-0.993587384	-2.311544229	
26	5502	100360.7379						Commercial1A_Balls	1	-1.084510704	-0.865016245	0.219494459	
27	5752	100860.7379						Commercial1A_Balls	1	0.667493255	-1.569089434	-2.236582689	
28	6002	101360.7379						Commercial1A_Balls	1	0.748584117	0.198285023	-0.550299094	
29	6252	101860.7379						Commercial1A_Balls	1	0.408910714	-0.500425304	-0.909336017	
30	6502	102360.7379						Commercial1A_Balls	1	-0.392368471	-0.608534641	-0.21616617	
31	6752	102860.7379						Commercial1A_Balls	1	0.824949529	1.057135528	0.232185998	
32	7002	103360.7379						Commercial1A_Balls	1	1.294804868	0.502393195	-0.792411673	
33	7252	103860.7379						Commercial1A_Balls	1	0.50952742	0.405197564	-0.104329856	
34	7502	104360.7379						Commercial1A_Balls	1	-1.130128001	0.040686147	1.170814148	
35	7752	104860.7379						Commercial1A_Balls	1	1.053718544	1.169417083	0.115698539	
36	8002	105360.7379						Commercial1A_Balls	1	0.941200786	-1.881122302	-2.822323087	
37	8252	105860.7379						Commercial1A_Balls	1	0.066641736	-0.663989388	-0.730631124	
38	8502	106360.7379						Commercial1A_Balls	1	-0.415353374	-0.504879235	-0.08952586	
39	8752	106860.7379						Commercial1A_Balls	1	0.301057403	0.292805975	-0.008251429	

Report

By clicking the "Reports" button it is also possible to retrieve an .html report with the R-code embedded and a step-by-step description of the algorithm. Here in the report, you will also find information/messages on troubleshooting if something is going wrong.

> **Note:**
>
> You can find this .html file only for individual FA R Notebook. There is no .html output for aggregated FA R Notebook.

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EEG Frontal Asymmetry

Code

Code
Code
Code
Code

Study: `SonyBravia_ET_FEA_EDA_EEG_EOG (1)`
Stimulus: `Commercial1A_Balls`
Respondent: `R4`
Sensor: `Enobio`
Left frontal electrode set: `Ch19`
Right frontal electrode set: `Ch10`

Code
Code
Code

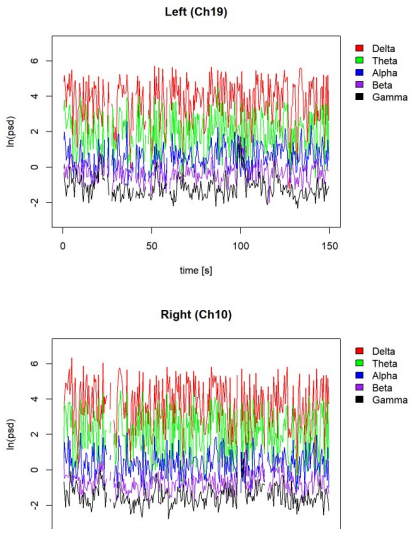
Based on the methods described in (Allen, Coan, and Nazarian 2004), this R notebook has performed the following steps on the raw data:

- Filtering of low and high frequencies (here frequencies between 0.5 - 100 Hz) using a zero phase-lag butterworth **band-pass** filter. The default value of 0.5 and 45 Hz have been chosen to avoid distortion of the signal but could be adapted based on the use case.
- **Notch filtering** of the power noise using a zero phase-lag butterworth filter (here filtering the 50 Hz frequency).
- **Artifact rejection** based on the selected threshold parameter (here 100 uV). Data points with an absolute value superior to this threshold are considered as outliers and removed for further analysis. As a quality check, the percentage of data removed is computed and displayed (high percentage means poor signal quality). The default threshold value (120 uV) has been chosen to preserve the typical EEG signal amplitude of an adult human (range from 10 to 100 uV). Depending on the range of the signal of interest, the parameter might need to be adapted.
- Computation of the **Power Spectral Density** (PSD) estimate using Fast Fourier Transform (FFT) (see here for a general description of FFT or here for a detailed math description). Resulting frequencies are in range [0, Fn], where Fn (0.5 * samplingRate) is Nyquist rate.
- In other words, the filtered signal is chunked into 1 second hamming windows (with an overlap of 0.5 second). If data points for a given 1 second window are invalid (i.e. considered as an artifact), the time serie is removed from the computation.
- **Power band** computation by averaging PSD estimate across frequencies of interest (Delta (1 - 3 Hz), Theta (4 - 7 Hz), Alpha (8 - 12 Hz), Beta (13 - 25 Hz) and Gamma (26 - 40 Hz).
- Computation of **Frontal Asymmetry** by computing the difference score in each selected powerband. For example, computation of Frontal Alpha Asymmetry: ln(Alpha_Right(Ch10)) - ln(Alpha_Left(Ch19)) (Briesemeister et al. 2013). Averaging the PSD estimates on each side and then do the substraction on natural log transformed values give the **mean frontal asymmetry** value.

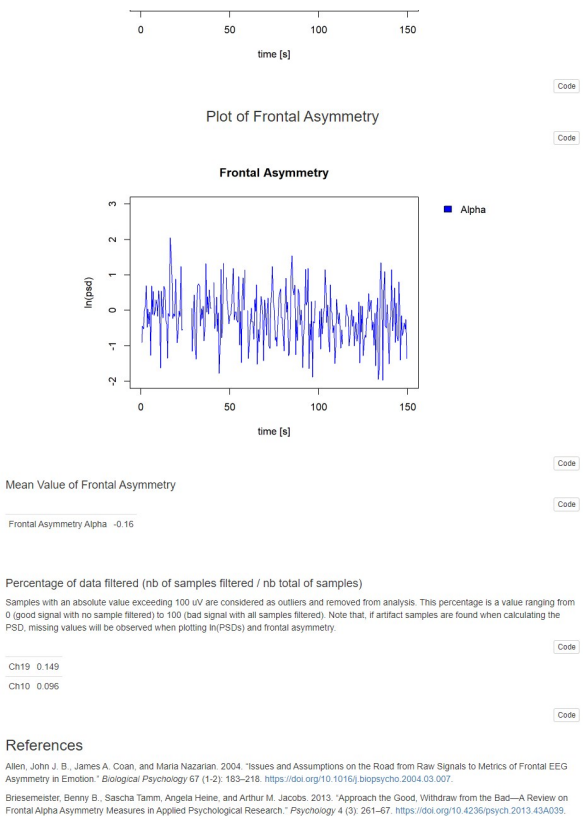
Code

Plots of ln(PSD) for the left/right electrode set

Code



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References

Briesemeister, B. B., Tamm, S., Heine, A., & Jacobs, A. M. (2013). Approach the good, withdraw from the bad—a review on frontal alpha asymmetry measures in applied psychological research. Psychology, 4(03), 261. Coan, J. A., Allen, J. J., & Harmon-Jones, E. (2001). Voluntary facial expression and hemispheric asymmetry over the frontal cortex. *Psychophysiology*, 38(6), 912-925.

Allen, J. J., Coan, J. A., & Nazarian, M. (2004). Issues and assumptions on the road from raw signals to metrics of frontal EEG asymmetry in emotion. *Biological psychology*, 67(1-2), 183-218.

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